

Experiment 1

Data Acquisition and Data Profiling using BI Tools

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Aim

To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to assess data quality, understand data structure, identify anomalies, and prepare data for further analytical processing.

Tools / Software Required

- Microsoft Power BI Desktop
- Sample datasets: Sales_Data.csv, Employee_DB.xlsx, and a sample SQL database

Theory

Business Intelligence (BI)

Business Intelligence refers to a set of technologies, processes, and practices that enable organizations to collect, integrate, analyse, and present business information. BI helps transform raw data into meaningful, actionable insights that support better decision-making across an organization.

Key components of a BI system include:

- Data Sources: Databases (SQL/NoSQL), flat files, cloud stores, and APIs
- ETL Pipeline: Extract, Transform, Load processes that move data from source to destination
- Data Warehouse / Data Lake: Centralized repositories for structured or unstructured data
- BI & Reporting Tool: Microsoft Power BI for data connection, transformation, and visualization
- Analytics Layer: Descriptive, diagnostic, predictive, and prescriptive analytics

Data Acquisition

Data Acquisition is the process of collecting and importing data from various heterogeneous sources into a unified environment for processing. It is the first and foundational step in any BI pipeline. In Power BI, data is acquired through the Get Data option, which supports a wide range of connectors including flat files (CSV, XLSX), relational databases (SQL Server, MySQL), cloud sources (SharePoint, Azure), and web-based sources.

Data Profiling

Data Profiling is the process of examining data from an existing source and collecting statistics, summaries, and metadata about that data. It is a critical quality assurance step that reveals the structure, content, and quality of a dataset before further processing. Data profiling covers the following key dimensions:

- Completeness: Percentage of non-null values in each column.
- Uniqueness: Ratio of distinct to total values — detects duplicate records.
- Validity: Whether values conform to defined rules or expected formats.
- Consistency: Uniformity of values across related datasets.
- Accuracy: Correctness of values relative to real-world facts.
- Timeliness: Currency of data relative to the reporting period.

Data Profiling in Power BI — Power Query Editor

Microsoft Power BI provides built-in data profiling capabilities through the Power Query Editor. Under the View tab, three profiling features can be enabled:

- Column Quality: Displays the percentage of valid, error, and empty values per column.
- Column Distribution: Shows a histogram of value frequency along with the distinct and unique count for each column.
- Column Profile: Provides detailed statistics for a selected column including minimum, maximum, mean, median, standard deviation, and a full value distribution chart.

Conduct

Part A: Data Acquisition

Step 1: Launch Power BI Desktop and import the CSV file.

Home > Get Data > Text/CSV
 Browse to Sales_Data.csv > Load

Step 2: Import the Excel file.

Home > Get Data > Excel Workbook
Browse to Employee_DB.xlsx > Select the appropriate sheet > Load

EmployeeID	Name	Department	Region	Salary
1 E001	Employee 1	IT	North	74.
2 E002	Employee 2	Finance	North	70
3 E003	Employee 3	IT	South	72.
4 E004	Employee 4	Finance	East	72.
5 E005	Employee 5	Sales	South	32.
6 E006	Employee 6	Sales	East	62.
7 E007	Employee 7	Sales	South	88.
8 E008	Employee 8	HR	West	59
9 E009	Employee 9	Finance	South	88.
10 E010	Employee 10	Sales	West	70.

Step 3: Connect to the SQL database and import the Transactions table.

Home > Get Data > SQL Server
Enter server name and database credentials > Select Transactions table > Load

Verify that all three data sources appear in the Data pane on the right-hand side of the Power BI window.

Part B: Data Profiling in Power Query Editor

Step 4: Open Power Query Editor.

Home > Transform Data

Navigators

Display Options ▾

Employee_DB.xlsx [1]

☒ Employees

Employees

EmployeeID	Name	Department	Region	Salary	JoiningDate
E001	Employee 1	IT	North	74226	28-0
E002	Employee 2	Finance	North	70602	25-0
E003	Employee 3	IT	South	72444	06-0
E004	Employee 4	Finance	East	72548	23-0
E005	Employee 5	Sales	South	32971	27-0
E006	Employee 6	Sales	East	62284	29-0
E007	Employee 7	Sales	South	88147	05-0
E008	Employee 8	HR	West	59729	15-0
E009	Employee 9	Finance	South	88026	01-0
E010	Employee 10	Sales	West	70347	22-0
E011	Employee 11	IT	North	75238	15-0
E012	Employee 12	HR	West	59143	01-0
E013	Employee 13	HR	East	36511	04-1
E014	Employee 14	IT	East	33968	12-0
E015	Employee 15	IT	South	38005	30-0
E016	Employee 16	Finance	North	73435	06-0
E017	Employee 17	Sales	North	81563	08-1
E018	Employee 18	HR	South	81575	02-0
E019	Employee 19	HR	North	84382	23-0
E020	Employee 20	HR	South	83284	10-0
E021	Employee 21	HR	East	80749	01-0
E022	Employee 22	Sales	East	86256	29-0
E023	Employee 23	HR	East	82331	28-0

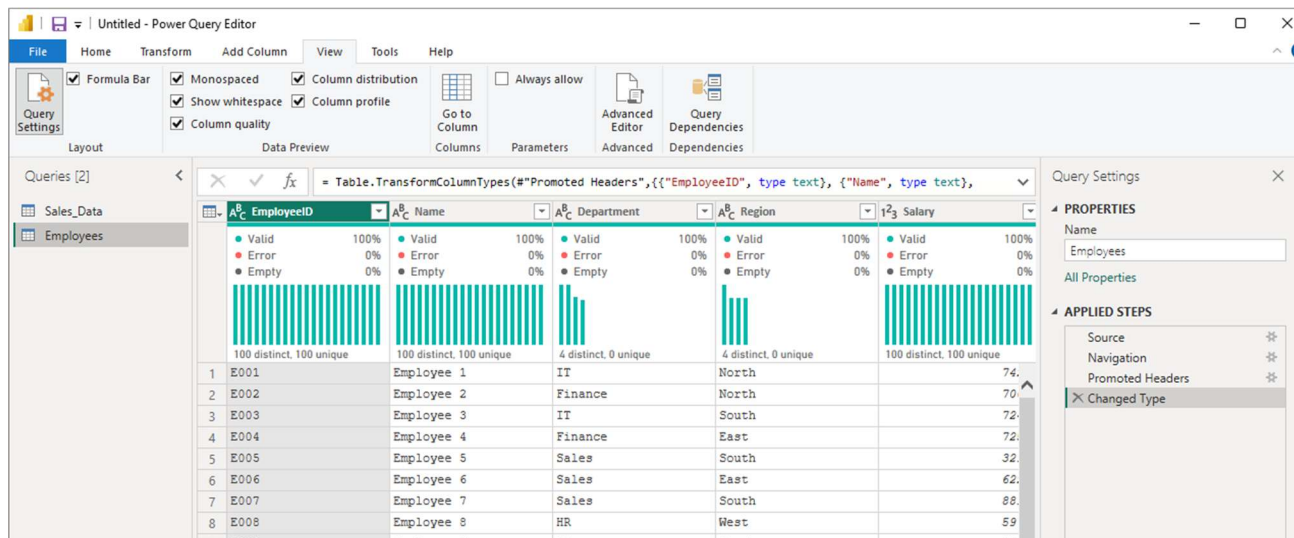
Load

Transform Data

Cancel

Step 5: Enable data profiling features.

```
View tab > tick: Column Quality
View tab > tick: Column Distribution
View tab > tick: Column Profile
```

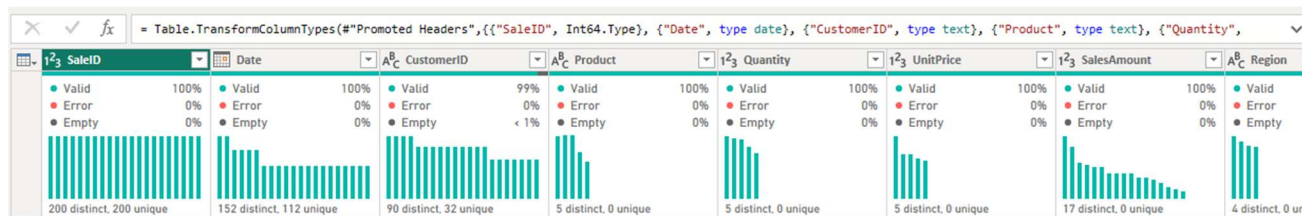


Select each column one at a time and observe the profiling statistics displayed below the data preview grid.

Step 6: Record profiling observations.

For each column in each dataset, note the following:

- Data type and total row count
- Null / empty percentage (from Column Quality)
- Distinct value count and unique value count (from Column Distribution)
- Minimum, maximum, mean, and standard deviation for numeric columns (from Column Profile)
- Any errors, mismatched data types, or unexpected values observed

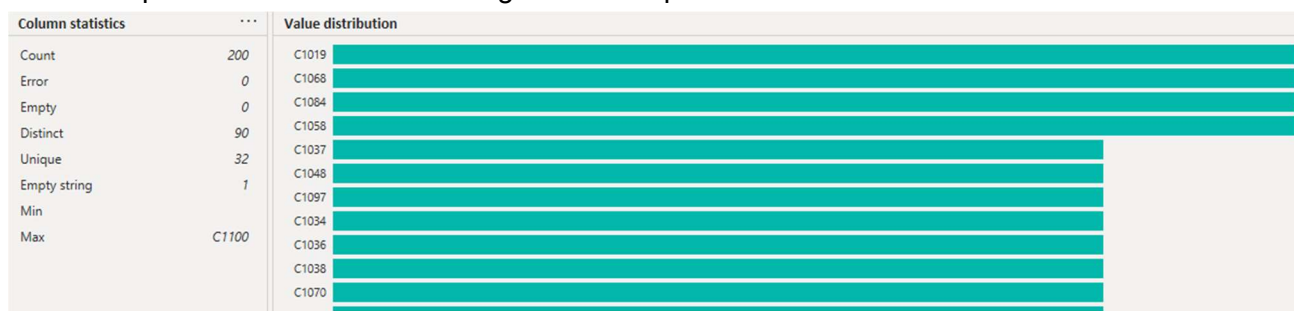


Part C: Identify and Document Data Quality Issues

Step 7: Based on the profiling results, identify data quality issues.

Look for the following common issues:

- High null percentage in important columns such as customer ID or date fields
- Columns with only one unique value — zero variance and no analytical value
- Numeric columns containing text values indicating a data type mismatch
- Date columns with future dates or dates outside the expected range
- Duplicate rows identified through a low uniqueness ratio



Document at least three data quality issues found and note the suggested remediation action for each.

Result

Data from multiple heterogeneous sources (CSV, Excel, and SQL database) was successfully acquired and imported into Microsoft Power BI Desktop. Data profiling was performed using the Power Query Editor's built-in tools — Column Quality, Column Distribution, and Column Profile. Key data quality dimensions including completeness, uniqueness, and validity were assessed across all datasets, and data quality issues were identified and documented for remediation prior to analysis.

Learning Outcomes

After completing this experiment, students will be able to:

- Explain the concept of Business Intelligence, its key components, and the role of data acquisition in a BI pipeline. (CO1 - BT2)
- Describe different data source types and the methods used to connect and import them into Power BI. (CO1 - BT2)
- Connect to multiple heterogeneous data sources and successfully import datasets using Power BI's Get Data feature. (CO2 - BT3)
- Apply data profiling techniques in Power Query Editor to assess data quality dimensions across real-world datasets. (CO2 - BT3)
- Interpret profiling statistics to identify data quality issues and propose suitable remediation strategies. (CO2 - BT3)

Text / Reference Books

- T - Sharda, Delen & Turban, Business Intelligence and Analytics: Systems for Decision Support, Pearson Education
- R - Cindi Howson, Successful Business Intelligence: Unlock the Value of BI & Big Data, McGraw-Hill
- Microsoft Power BI Official Documentation: docs.microsoft.com/power-bi